

Theoretical Computer Science Exam, January 22, 2024

The exam consists of **4 exercises**. Available time: 1 hr 30 min.; you may write your answers in Italian, if you wish.

All exercises must be addressed for the written exam to be considered sufficient.

F.NAME L.NAME PERSON CODE

Exercise 1 (8 pts)

Consider the language L generated by the following grammar (axiom is S)

$$S \rightarrow B S a \mid C$$

$$B C \rightarrow C b b$$

$$C \rightarrow c$$

Answer the following questions

- Write a derivation for the 3 shortest strings of language L .
- Define precisely language L . Preferably use a set expression of the type $L = \{ \dots \mid \dots \text{predicate } \dots \}$.
- What is the least powerful type of automaton that accepts L ?
- Design a minimal power automaton that accepts L .

Exercise 2 (8 pts)

- Specify, by means of a sentence of Monadic First-Order logic (or Second-Order only if necessary) the language L_1 over the alphabet $\{a, b, c\}$ defined by the following condition. There are two letters b occurring at distance ≥ 2 such that between them there are only letters c . For instance, $cabccbbba \in L_1$, while $acbacbbba \notin L_1$.
- Specify, by means of a sentence of Monadic First-Order logic (or Second-Order only if necessary) the language L_2 over the alphabet $\{a, b, c\}$ defined by the following condition. There are two letters a respectively occurring at an even position and at the next even position. For instance, $bacac \in L_2$, while $baabca \notin L_2$.

Exercise 3 (8 pts)

Consider the function $g(x)$ defined as follows

$$g(x) = \begin{cases} 1 & \text{if } \neg \exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z)) \\ 0 & \text{otherwise} \end{cases}$$

and say if function g is computable. Provide a suitable justification for your answer.

Next consider the set S defined as follows

$$S = \{ x \mid \neg \exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z)) \}$$

and say if set S is decidable. Provide a suitable justification for your answer.

Say if set S is semidecidable. Provide a suitable justification for your answer.

Consider next the function $g'(x)$ defined as follows

$$g'(x) = \begin{cases} 1 & \text{if } \exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z)) \\ \perp & \text{otherwise} \end{cases}$$

and say if function g' is computable. Provide a suitable justification for your answer.

Exercise 4 (8 pts)

Consider the following language L , over the alphabet $\{a, b, c\}$,

$$L = \{ x c y \mid x, y \in \{a, b\}^+ \wedge \#a(x) > \#a(y) \wedge \#b(x) < \#b(y) \}$$

where, for any string s , $\#a(s)$ (respectively, $\#b(s)$) denotes the number of occurrences of letter a (respectively, letter b) in string s . For instance, $aabbacababb \in L$, while $babbacabbaa \notin L$.

Describe informally but precisely a k -tape Turing machine that recognizes L and determine its time and space complexity class. Suitably motivate your answers.

Describe informally but precisely a RAM machine that recognizes L and determine its time and space complexity class, both using the constant and the logarithmic cost criterion. Suitably motivate your answers.

Solutions

Exercise 1

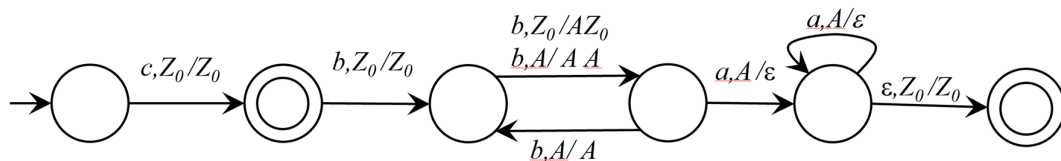
a) $S \Rightarrow C \Rightarrow c$

$$S \Rightarrow B S a \Rightarrow B C a \Rightarrow C b b a \Rightarrow c b b a = c b^2 a$$

$$S \Rightarrow B S a \Rightarrow B B S a a \Rightarrow B B C a a \Rightarrow B C b b a a \Rightarrow C b b b b a a \Rightarrow c b b b b a a = c b^4 a^2$$

b) $L = \{ c b^{2n} a^n \mid n \geq 0 \}$

c) The least powerful type of automaton that accepts L is a deterministic pushdown automaton.



d)

Exercise 2

$$\exists x \exists y (y \geq x + 2 \wedge b(x) \wedge b(y) \wedge \forall z (x < z < y \rightarrow c(z)))$$

$$\exists X (\neg X(0) \wedge \forall x (\neg last(x) \rightarrow (X(x) \leftrightarrow \neg X(x+1)))) \wedge$$

$$\exists y (X(y) \wedge a(y) \wedge a(y+2))$$

)

Exercise 3

Question Consider the function $g(x)$ defined as follows

$$g(x) = \begin{cases} 1 & \text{if } \neg \exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z)) \\ 0 & \text{otherwise} \end{cases}$$

and say if function g is computable. Provide a suitable justification for your answer.

Answer Function g is the characteristic predicate of the set

$$S = \{ x \mid \neg \exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z)) \}$$

This is the set of indices of TMs that compute functions in the set

$$F = \{ f \mid \neg \exists y \exists z (f(y) \neq \perp \wedge f(z) \neq \perp \wedge f(y) \neq f(z)) \}$$

i.e., functions that are constant where they are defined. The set F satisfies the hypothesis of the Rice theorem, therefore function g is not computable.

Question Next consider the set S defined as follows

$$S = \{ x \mid \neg \exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z)) \}$$

and say if set S is decidable. Provide a suitable justification for your answer.

Answer The set S is not decidable, because it is the set of indices of TMs that compute functions in the set F above.

Question Say if set S is semidecidable. Provide a suitable justification for your answer.

Answer Consider the complement S' of set S .

$$S' = \{ x \mid \exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z)) \}$$

Set S' is semidecidable by (dovetailing) simulation: given a value x , the simulation looks for two distinct values y and z such that $f_x(y) \neq f_x(z)$. If these two values exist then the simulation will eventually find them.

As a consequence, the set S , which is undecidable, cannot be semidecidable, otherwise both S and S' would be semidecidable and hence decidable.

Question Consider next the function $g'(x)$ defined as follows

$$g'(x) = \begin{cases} 1 & \text{if } \exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z)) \\ \perp & \text{otherwise} \end{cases}$$

and say if function g' is computable. Provide a suitable justification for your answer.

Answer The function g' is in fact computable. If it is the case that $\exists y \exists z (f_x(y) \neq \perp \wedge f_x(z) \neq \perp \wedge f_x(y) \neq f_x(z))$, then a suitable Turing machine can verify this condition by carrying out a simulation (using the dovetail technique) and then output one; if the condition does not hold, then the same Turing machine would never terminate, therefore it would “compute” the value $g'(x) = \perp$.

Exercise 4

A Turing machine with two memory tapes T1 and T2 reads the first part x of the input string and stores in unary $\#a(x)$ on the first tape and $\#b(x)$ on the second tape, e.g. by writing a mark '*' on T1 when it reads an a and on T2 when it reads a b . When it reads the c it moves both memory heads left on the last written '*' mark. Then it reads the second part y while it moves left the head of T1 if it reads an a and the head of T2 if it reads a b . When the input string is completely read it is accepted iff the head of T1 has not reached the left end of the T1 and the head of T2 has reached it. The time complexity class is $\Theta(n)$ (where n is the length of the input) because the input head always moves right and the string is accepted or rejected immediately after reading its last symbol. The space complexity class is $\Theta(n)$ because a number of symbols equal to $|x|$ is stored in the memory tapes, and in the worst case $|x|$ is much greater than $|y|$.

A RAM machine uses four variables for counting $\#a(x)$, $\#a(y)$, $\#b(x)$ and $\#b(y)$, initializing them to 0 and increasing them in the obvious way while it reads the input string. After completing the reading of the input it compares the four counters to determine if the condition characterizing the language is satisfied. With the constant cost criterion the time complexity class is $\Theta(n)$ for the same reason as for the Turing machine, and the space complexity is constant, because only the four counters are stored. With the logarithmic cost criterion the time complexity is $\Theta(n \log n)$ because of the cost of increasing the counters and the space complexity is $\Theta(\log n)$ because of the binary encoding of the value of counters, which is $\Theta(n)$.